

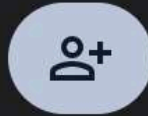
Lab Workshop on Embedded C and ARM Cortex Microcontrollers

Day2: Lab Experiments: C Programming

1. Write a program to print "NIELIT" in the console
2. Write a program to enter two numbers through keyboard and print their sum
3. Two numbers are input through the keyboard into two locations C and D. Write a program to interchange the contents of C and D.
4. An integer is input through the keyboard. Write a program to find out whether it is an odd number or even number.
5. Write a program to print the size of int, longint, double,char,float.

Note: write all the programs using text editor (nano or vi) and o compile using terminal.

× prog1.c



```
#include <stdio.h>
```

```
int main()
```

```
{
```

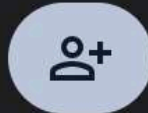
```
    printf(" NIELIT \n");
```

```
    return 0;
```

```
}
```



× prog2.c



```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int num1, num2;
```

```
    printf("Enter 2 integers:");
```

```
    scanf("%d %d", &num1,&num2);
```

```
    printf("The sum of two integers is %d\n", num1+num2 );
```

```
    return 0;
```

```
}
```



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Tab 1



```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int C, D;
```

```
    printf("Enter 2 integers:");
```

```
    scanf("%d %d", &C,&D);
```

```
    printf("Values before swapping are C =  
%d , D = %d \n", C, D );
```

```
    // using third variable
```

```
    int temp;
```

```
    temp = C;
```

```
    C = D;
```

```
    D = temp;
```

```
    printf("Values after swapping with 3rd  
variable are C = %d , D = %d \n", C, D );
```

```
    // without third variable
```

```
    C = C+D;
```

```
    D = C-D;
```

```
    C = C-D;
```

```
    printf("Values again swapping with out  
3rd variable are C = %d , D = %d \n", C, D  
);
```

```
    return 0;
```

```
}
```

× prog4.c



```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int num;
```

```
    printf("Enter the integer to check odd or  
even:");
```

```
    scanf("%d", &num);
```

```
    if(num % 2 == 0)
```

```
        printf("%d is even \n", num);
```

```
    else
```

```
        printf("%d is odd \n", num);
```

```
    return 0;
```

```
}
```



× prog5.c



```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    printf("Size of int datatype is %lu\n",sizeof(int));
```

```
    printf("Size of int datatype is %lu\n",sizeof(long int));
```

```
    printf("Size of int datatype is %lu\n",sizeof(double));
```

```
    printf("Size of int datatype is %lu\n",sizeof(char));
```

```
    printf("Size of int datatype is %lu\n",sizeof(float));
```

```
    return 0;
```

```
}
```



Tab 1

